

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT

Magnesium sulfate / DEMO 50% w/v concentrate for solution for injection or infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 10 mL ampoule contains 5 g magnesium sulfate heptahydrate (500 mg/mL).

1 mL contains 2.03 mmol (49.3 mg = 4.06 mEq) magnesium anions.

Each ampoule contains 5 g of magnesium sulfate heptahydrate, 493 mg or 20.3 mmol (40.6 mEq) of elemental magnesium.

Clear, colourless solution, free from visible particles.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection or concentrate for solution for injection or infusion, with a pH of between 5.5 and 7 and osmolality of 2000 mOsm/kg approximately.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Parenteral administration of magnesium is indicated in:

- treatment and prophylaxis of acute hypomagnesemia.
- prevent hypomagnesemia in patients receiving total parenteral nutrition.
- prevention and treatment of life-threatening seizures in the treatment of toxemias of pregnancy (pre-eclampsia and eclampsia).

4.2 Posology and method of administration

Magnesium sulfate 50% w/v solution may be administered by the intravenous (preferred method) or intramuscular (painful, avoid if possible) routes (see below for method of administration and section 4.4).

Posology

The dose of magnesium should be adjusted according to the patient's individual requirements and response.

The total adult daily dose should not exceed 30 to 40 g of magnesium sulfate per day.

Mild Hypomagnesemia:

Adult: A dose of 1g magnesium sulfate (8 mEq) intramuscularly every 6 hours for 4 doses is recommended.

Severe Hypomagnesemia:

Adult: A dose of 0.25 g/kg magnesium sulfate intramuscularly over 4 hours is recommended. Alternatively, a dose of 5 g may be given by slow intravenous infusion over 3 hours.

For Total Parenteral Nutrition use:

Adults: A dose of 0.5 to 3.0 g magnesium sulfate (4 to 24 mEq) daily may be administered.

Infants: A dose of 0.25 to 1.25 g magnesium sulfate (2 to 10mEq) daily may be administered.

Toxaemia of Pregnancy

An initial intravenous dose of 4 g magnesium sulfate is recommended. This is followed by an intramuscular dose of 4 to 5 g into each buttock. This may be followed by a dose of 4 to 5 g into alternate buttocks every four hours as needed. Alternatively, the initial IV dose may be followed by an infusion of 1 to 2 g/hr.

Geriatrics

Geriatric patients often require reduced dosage because of impaired renal function. In patients with severe impairment, dosage should not exceed 20 g in 48 hours. Serum magnesium should be monitored in such patients.

Method of administration

Intravenous use in adults and adolescents

Intravenous infusion: For the intravenous route, the 50% w/v solution requires dilution to a concentration of not more than 20% (≤ 200 mg/mL magnesium sulfate heptahydrate) – with a suitable diluent, such as Glucose 5% or sodium chloride 0.9%. Infuse via a volumetric infusion device at a rate appropriate to the indication (see posology above).

Intravenous injection: Give by slow IV injection at a rate appropriate to the indication (see posology above).

Intravenous use in children:

Rate of administration should not exceed 0.02 mL/kg/min of appropriately diluted Magnesium sulfate 50% w/v solution (corresponding to 0.04 mmol/kg/min $\approx$ 0.001 g/kg/min Mg^{2+}).

Deep Intramuscular injection (adults only)

For the intramuscular route, the 50% w/v solution should be used undiluted or diluted to 25% w/v. If the total dose to be administered exceeds 5 mL, the injection volume should be divided between more than one deep muscular injection site.

Compatibilities: Magnesium sulfate 50% w/v admixtures with Sodium chloride 9 mg/mL solution for injection and Dextrose 50 mg/mL solution for injection, is stable for 48 hours at $25 \pm 2^\circ C$ and at $5 \pm 3^\circ C$ at a concentration of 20% (equal to 200 mg/mL).

4.3 Contraindications

Hypersensitivity to magnesium and its salts or to any of the excipients listed in section 6.1.
Hepatic encephalopathy, hepatic failure, renal failure.
Severe renal impairment (glomerular filtration rate under 25 mL/h), anuria.
Parenteral administration of the medicinal product is contraindicated in patients with heart block (class I-III) or myocardial damage and myasthenia gravis.

Magnesium sulfate should not be administered to pregnant women in the two hours prior to delivery, unless it is the only therapy available to prevent eclamptic seizures. There is a risk that the neonate will be born with hypermagnesemia and depressed breathing.

4.4 Special warnings and precautions for use

Magnesium sulfate heptahydrate may precipitate an acute myasthenic crisis. Sensitivity to parenteral magnesium has been reported.

An intravenous preparation of a calcium salt (eg calcium gluconate) should be readily available for use when magnesium sulfate heptahydrate is given intravenously.

Use in renal impairment

Magnesium should be administered with caution in patients with impaired renal function, since the risk of hypermagnesaemia is increased in these patients.

Use in the elderly

No data available.

Paediatric use

No data available.

Effects on laboratory tests

Monitoring of serum magnesium levels is advised at periodic intervals during therapy to ensure that normal serum magnesium levels are not exceeded.

The patellar reflex should be tested prior to administering repeat doses of magnesium sulfate heptahydrate. Suppression of the reflex is an indication of magnesium intoxication.

Respiration rate should be determined and should be at least 16 per minute prior to each dose of magnesium sulfate heptahydrate, as respiratory depression is the most critical side effect of the medication.

Urine output should be monitored and should be at least 100 mL during the four hours preceding dosing, to ensure adequate excretion of magnesium.

4.5 Interaction with other medicinal products and other forms of interaction

Cardiac glycosides/digitalis: Magnesium salts should be administered with caution in patients treated with cardiac glycosides, since heart block may occur if calcium salts are required to treat magnesium toxicity. (see **Section 4.9 Overdose**)

CNS depressants: Concurrent use of magnesium salts and CNS depressant drugs may result in an enhanced CNS depressant effect.

Neuromuscular blocking agents: Concurrent use of magnesium salts with neuromuscular blocking agents may result in an excessive neuromuscular blockade.

Nifedipine: Concurrent use of magnesium sulfate heptahydrate and nifedipine may result in an exaggerated hypotensive response.

Also refer to **Section 6.2 Incompatibilities**.

4.6 Fertility, pregnancy and lactation

Effects on fertility

No data available.

Use in pregnancy

Magnesium sulfate heptahydrate is administered to pregnant women to treat seizures associated with severe pre-eclampsia and eclampsia. Magnesium sulfate heptahydrate readily crosses the placenta. Foetal serum concentrations approximate those of the mother. If magnesium sulfate heptahydrate is administered in the two hours preceding delivery, the neonate may be born with signs of hypermagnesaemia, including respiratory depression, and therefore DBL® Magnesium Sulfate Concentrated Injection should not be given in the two hours preceding delivery unless it is the only therapy available to prevent or treat eclamptic seizures.

Bony abnormalities and congenital rickets have been reported in neonates born to mothers treated with parenteral magnesium sulfate heptahydrate for prolonged periods of time (5-7 days duration).

Use in lactation

After intravenous administration, magnesium is distributed into breast milk, and the concentration of magnesium in the breast milk is approximately twice that in the maternal serum. Magnesium salts should therefore be used with caution in lactating patients. However, magnesium is cleared from the breast milk within 24 hours of the cessation of the infusion.

4.7 Effects on ability to drive and use machines

The effects of this medicine on a person's ability to drive and use machines were not assessed as part of its registration.

4.8 Undesirable effects

Excessive administration of magnesium sulfate heptahydrate may result in hypermagnesaemia. The signs of hypermagnesaemia may include: nausea, vomiting, flushing, hypotension, muscle weakness, muscle paralysis, blurred or double vision, CNS depression and loss of reflexes. More severe hypermagnesaemia may result in respiratory depression, respiratory paralysis, renal failure, coma, cardiac arrhythmias and cardiac arrest. Hypocalcaemia with tetany, secondary to hypermagnesaemia, has been reported.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the National Pharmaceutical Regulatory Agency (NPRA) Adverse Drug Reaction Centre by visiting the website www.npra.gov.my.

4.9 Overdose

Clinical features

Hypermagnesaemia may occur when large doses of magnesium are given, especially in patients with renal failure. Signs of hypermagnesaemia include: nausea, vomiting, flushing, hypotension, muscle weakness, muscle paralysis, blurred or double vision, CNS depression and loss of reflexes. More severe hypermagnesaemia may result in respiratory depression, respiratory paralysis, renal failure, coma, cardiac arrhythmias and cardiac arrest.

Treatment

In the treatment of hypermagnesaemia, the following measures may be required:

- blood pressure and respiratory support
- intravenous administration of 2.5 to 10 mmol calcium salts (such as calcium gluconate) reverses the effects of magnesium toxicity
- dialysis may be required, particularly if renal function is impaired
- if renal function is normal, adequate fluids should be given so that urine output is at least 60 mL/hr to assist removal of magnesium from the body.
- physostigmine (0.5 to 1.0 mg subcutaneously) may be helpful, but routine use is not recommended due to the potential toxicity.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Other mineral supplements, magnesium sulfate

ATC code: A12CC02

Magnesium is the second most abundant cation in intracellular fluid and is an essential body electrolyte.

The body contains about 25 g of magnesium (about 14 mmol per kg body weight), approximately 60% of which is found in the skeleton. The daily amount of magnesium required by an adult is of the order of 270 to 350 mg (about 11 to 14 mmol). Symptomatic hypomagnesaemia is associated with a deficit of 0.5 – 1.0 mmol/kg.

Mechanism of action

Magnesium is the second most abundant cation of intracellular fluid. It is an essential cation in over 300 enzymatic processes, and is necessary for several steps in glycolysis, the Krebs cycle and in protein and nucleic acid synthesis. It is thus vital for normal energy storage and transfer.

Magnesium plays an important role in neurochemical transmission, and is essential for proper neurochemical functioning.

Magnesium has an anticonvulsant effect. It possibly has antiarrhythmic effects and a role in calcium homeostasis and bone mineralisation. There is conflicting evidence that the routine use of intravenous magnesium sulfate heptahydrate in the setting of acute myocardial infarction is beneficial. Deficiency of magnesium is closely associated with other electrolyte disturbances, particularly hypocalcaemia and hypokalaemia. The specific symptoms of hypomagnesaemia are therefore difficult to determine, but may include nausea, vomiting, muscle weakness, neuromuscular dysfunction such as paraesthesia, tremor and cramp, tachycardia and cardiac arrhythmias.

Clinical efficacy and safety

The normal concentration of magnesium in plasma is around 0.65 to 1.0 mmol/L. Serum magnesium levels in the range 1.5 – 2.5 mmol/L cause vasodilatation in the peripheral and coronary circulation and corresponding increases of 20 – 25% in cardiac output and coronary blood flow. There is little change in heart rate or blood pressure. Animal studies suggest that the effect of magnesium ions on cardiac muscle is to slow the rate of the sinoatrial node impulse formation and prolong conduction time. Limited data on patients with no evidence of heart disease indicate that intravenous magnesium prolongs PR interval, H (atria-His bundle) interval, antegrade AV nodal effective refractory period and sinoatrial conduction time. Within this concentration range there are no detectable effects on CNS function or neuromuscular transmission.

When given intravenously, magnesium sulfate has an immediate onset of action and its duration of activity is about 30 minutes. The onset of action of intramuscular magnesium sulfate is about one hour and its duration of action 3 – 4 hours.

5.2 Pharmacokinetic properties

The 95% confidence intervals for magnesium levels in healthy Australian subjects are: neonate 0.6 to 0.9 mmol/L and adult 0.8 to 1.0 mmol/L.

Approximately 50% of magnesium in the body is found in bone, with the majority of the remainder stored in muscle and soft tissue. 1% or less is contained in the extracellular compartment, of which approximately 33% is protein-bound, with a further 12% bound to anions.

Magnesium is primarily excreted in the urine, with small amounts excreted in faeces, saliva and breast milk. Over 90% of magnesium filtered by the kidneys is reabsorbed, mainly in the ascending limb of the Loop of Henle, with significant amounts also absorbed in the proximal and distal tubules. The clearance is proportional to the plasma magnesium concentration and the glomerular filtration rate. The onset of action after intramuscular injection is about 1 hour and after intravenous injection is nearly immediate. The duration of action after intramuscular injection is 3 to 4 hours, and after intravenous injection is about 30 minutes.

5.3 Preclinical safety data

There are no preclinical data of relevance to the prescriber additional to those already included in other sections of the SPC.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Water for injections

Sulfuric acid (for pH adjustment)

6.2 Incompatibilities

This medicinal product must not be mixed with other medicinal products/diluents except Glucose 5% or sodium chloride 0.9%.

Magnesium sulfate is incompatible with alkali hydroxides (forming insoluble magnesium hydroxide), alkali carbonates (forming insoluble magnesium carbonate) and salicylates. The activities of streptomycin sulfate and tetracycline sulfate are inhibited by magnesium ions.

6.3 Shelf life

36 months

After opening/dilution:

Chemical and physical in-use stability has been demonstrated for 48 hours at $25 \pm 2^{\circ}\text{C}$ and $5 \pm 3^{\circ}\text{C}$ when diluted to a concentration of not more than 200 mg/mL magnesium sulfate heptahydrate in Sodium chloride 0.9% or Glucose 5%.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibilities of the user and would normally not be longer than 48 hours at $25 \pm 2^{\circ}\text{C}$ and $5 \pm 3^{\circ}\text{C}$ unless reconstitution/dilution (etc.) has taken place in controlled and validated aseptic conditions.

6.4 Special precautions for storage

Store below 30°C .

After opening, the product should be used immediately.

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6.5 Nature and contents of container

Polypropylene ampoules of 10 mL containing a 50 % w/v solution of magnesium sulfate as heptahydrate for injection. Packed in cartons to contain 10, 20 or 50 ampoules.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

For intramuscular use, a 50% w/v solution is used. For intravenous use, the 50% w/v solution must be diluted before use, with a suitable diluent, such as Glucose 5% or sodium chloride 0.9%.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

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7. PRODUCT REGISTRATION HOLDER

Pahang Pharmacy Sdn. Bhd., Lot 5979, Jalan Teratai,
5 ½ Mile Off Jalan Meru, 41050 Klang, Selangor, Malaysia

8. PRODUCT REGISTRATION NUMBER(S)

<[To be completed nationally]>

9. MANUFACTURER

DEMO S.A., Pharmaceutical Industry, 21st km National Road Athens-Lamia, 14568 Krioneri, Attiki, Greece, T: +30 210 8161802, F: +30 210 8161587.

10. DATE OF REVISION OF THE TEXT

08/02/2023

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